UART BLUETOOTH CAR CONTROLLER

STATIC TIMING ANALYSIS - SELECTED REPORTS

OpenLane run: fix3 (re-run after fixing the max-fanout violation)

Fix applied: config.json "MAX_FANOUT_CONSTRAINT": 16

Previous run: clkbuf_2_2__f_clk/X fanout 14 vs limit 10 in all corners (now fixed)

Corner naming used in this report

FF = Fastest corner, TT = Typical corner, SS = Slowest corner

Each is analysed under 3 RC extractions: min, max, nom (9 combinations)

Signoff summary (all values from the fix3 reports)

RC	Setup WS	Hold WS	TNS	WNS	Slew	Fanout	Cap	FAIL
RC min	2.86	0.10	0.00	0.00	0	0	0	0
RC max	2.80	0.11	0.00	0.00	0	0	0	0
RC nom	2.83	0.11	0.00	0.00	0	0	0	0

Slew/Fanout/Cap = violation counts. FAIL = failing check entries in the report.

Total failing entries in this document: 0

Note on "No paths found"

The check 'report_checks --slack_max -0.01' lists paths with negative slack. 'No paths found.' there means no setup/hold path is failing in FF, TT or SS. It is the expected result for a clean design.

UART BLUETOOTH CAR CONTROLLER
STATIC TIMING ANALYSIS - SELECTED REPORTS
OpenLane run: fix3
==
SOURCE REPORT: designs/uart_bt_car/runs/fix3/reports/signoff/26-mca/rcx_min_sta.checks.rpt
==

```
========================================================================
report_checks -unconstrained
========================================================================
====================== FF Corner (Fastest) ==============================

Startpoint: reset (input port clocked by clk)
Endpoint: _412_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest (FF)

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          2.00    2.00 v input external delay
                   0.01   0.00    2.00 v reset (in)
     1    0.00                           reset (net)
                   0.01   0.00    2.00 v input2/A (sky130_fd_sc_hd__buf_1)
                   0.03   0.06    2.07 v input2/X (sky130_fd_sc_hd__buf_1)
     1    0.01                           net2 (net)
                   0.03   0.00    2.07 v _365_/A (sky130_fd_sc_hd__buf_8)
                   0.06   0.12    2.18 v _365_/X (sky130_fd_sc_hd__buf_8)
    14    0.10                           _203_ (net)
                   0.06   0.01    2.19 v _366_/A (sky130_fd_sc_hd__buf_8)
                   0.06   0.13    2.32 v _366_/X (sky130_fd_sc_hd__buf_8)
    16    0.09                           _204_ (net)
                   0.06   0.00    2.32 v _367_/A (sky130_fd_sc_hd__inv_2)
                   0.02   0.04    2.37 ^ _367_/Y (sky130_fd_sc_hd__inv_2)
     1    0.00                           _005_ (net)
                   0.02   0.00    2.37 ^ _412_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  2.37   data arrival time

                         10.00   10.00   clock clk (rise edge)
                          0.00   10.00   clock source latency
                   0.05   0.03   10.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.05   0.00   10.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.05   0.10   10.13 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.05   0.00   10.13 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.09   10.22 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                           clknet_2_3__leaf_clk (net)
                   0.04   0.00   10.22 ^ _412_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25    9.97   clock uncertainty
                          0.00    9.97   clock reconvergence pessimism
                          0.09   10.06   library recovery time
                                 10.06   data required time
---------------------------------------------------------------------------
                                 10.06   data required time
                                 -2.37   data arrival time
---------------------------------------------------------------------------
                                  7.69   slack (MET)


Startpoint: _454_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: motor_left_b (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest (FF)

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.05   0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.05   0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.05   0.11    0.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.05   0.00    0.15 ^ clkbuf_2_2__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.05   0.11    0.25 ^ clkbuf_2_2__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
    13    0.04                           clknet_2_2__leaf_clk (net)
                   0.05   0.00    0.25 ^ _454_/CLK (sky130_fd_sc_hd__dfrtp_1)
                   0.05   0.27    0.52 v _454_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.01                           net4 (net)
                   0.05   0.00    0.52 v output4/A (sky130_fd_sc_hd__clkbuf_4)
                   0.06   0.14    0.66 v output4/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           motor_left_b (net)
                   0.06   0.00    0.66 v motor_left_b (out)
                                  0.66   data arrival time

                         10.00   10.00   clock clk (rise edge)
                          0.00   10.00   clock network delay (propagated)
                         -0.25    9.75   clock uncertainty
                          0.00    9.75   clock reconvergence pessimism
```

```
                        -2.00    7.75   output external delay
                                 7.75   data required time

-------------------------------------------------------------------------------
                                 7.75   data required time
                                -0.66   data arrival time
-------------------------------------------------------------------------------
                                 7.09   slack (MET)



======================= SS Corner (Slowest) ==============================

Startpoint: reset (input port clocked by clk)
Endpoint: _412_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest (SS)

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          2.00    2.00 v input external delay
                   0.02   0.01    2.01 v reset (in)
     1    0.00                           reset (net)
                   0.02   0.00    2.01 v input2/A (sky130_fd_sc_hd__buf_1)
                   0.09   0.18    2.19 v input2/X (sky130_fd_sc_hd__buf_1)
     1    0.01                           net2 (net)
                   0.09   0.00    2.19 v _365_/A (sky130_fd_sc_hd__buf_8)
                   0.13   0.32    2.51 v _365_/X (sky130_fd_sc_hd__buf_8)
    14    0.10                           _203_ (net)
                   0.13   0.01    2.52 v _366_/A (sky130_fd_sc_hd__buf_8)
                   0.13   0.34    2.86 v _366_/X (sky130_fd_sc_hd__buf_8)
    16    0.09                           _204_ (net)
                   0.13   0.00    2.86 v _367_/A (sky130_fd_sc_hd__inv_2)
                   0.05   0.10    2.96 ^ _367_/Y (sky130_fd_sc_hd__inv_2)
     1    0.00                           _005_ (net)
                   0.05   0.00    2.96 ^ _412_/SET_B (sky130_fd_sc_hd__dfstp_1)
                          2.96   data arrival time

                         10.00   10.00   clock clk (rise edge)
                          0.00   10.00   clock source latency
                   0.11   0.07   10.07 ^ clk (in)
     1    0.01                           clk (net)
                   0.11   0.00   10.07 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.09   0.25   10.32 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.09   0.00   10.32 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.08   0.23   10.56 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                           clknet_2_3__leaf_clk (net)
                   0.08   0.00   10.56 ^ _412_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   10.31   clock uncertainty
                          0.00   10.31   clock reconvergence pessimism
                          0.27   10.58   library recovery time
                                 10.58   data required time
-------------------------------------------------------------------------------
                                 10.58   data required time
                                 -2.96   data arrival time
-------------------------------------------------------------------------------
                                  7.62   slack (MET)


Startpoint: _436_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _446_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest (SS)

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.11   0.07    0.07 ^ clk (in)
     1    0.01                           clk (net)
                   0.11   0.00    0.07 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.09   0.28    0.36 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.09   0.00    0.36 ^ clkbuf_2_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.10   0.28    0.63 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
    13    0.04                           clknet_2_0__leaf_clk (net)
                   0.10   0.00    0.63 ^ _436_/CLK (sky130_fd_sc_hd__dfrtp_2)
                   0.14   0.94    1.57 v _436_/Q (sky130_fd_sc_hd__dfrtp_2)
     6    0.02                           RX.count[9] (net)
                   0.14   0.00    1.57 v _223_/A (sky130_fd_sc_hd__or4_1)
                   0.25   1.41    2.98 v _223_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _102_ (net)
                   0.25   0.00    2.98 v _225_/A3 (sky130_fd_sc_hd__o41a_1)
                   0.22   0.95    3.93 v _225_/X (sky130_fd_sc_hd__o41a_1)
     4    0.01                           _104_ (net)
                   0.22   0.00    3.93 v _245_/B (sky130_fd_sc_hd__or4_1)
                   0.24   1.38    5.31 v _245_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _120_ (net)
```

```
                0.24    0.00    5.31 v _252_/A (sky130_fd_sc_hd__or3b_1)
                0.15    0.97    6.28 v _252_/X (sky130_fd_sc_hd__or3b_1)
     1    0.00                        _125_ (net)
                0.15    0.00    6.28 v _253_/S (sky130_fd_sc_hd__mux2_1)
                0.11    0.75    7.03 v _253_/X (sky130_fd_sc_hd__mux2_1)

     1    0.00                        _126_ (net)
                0.11    0.00    7.03 v _254_/A (sky130_fd_sc_hd__clkbuf_1)
                0.05    0.18    7.21 v _254_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                        _062_ (net)
                0.05    0.00    7.21 v _446_/D (sky130_fd_sc_hd__dfrtp_1)
                        7.21    data arrival time

                       10.00   10.00   clock clk (rise edge)
                        0.00   10.00   clock source latency
                0.11    0.07   10.07 ^ clk (in)
     1    0.01                        clk (net)
                0.11    0.00   10.07 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.09    0.25   10.32 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                        clknet_0_clk (net)
                0.09    0.00   10.32 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.08    0.23   10.56 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                        clknet_2_3__leaf_clk (net)
                0.08    0.00   10.56 ^ _446_/CLK (sky130_fd_sc_hd__dfrtp_1)
                       -0.25   10.31   clock uncertainty
                        0.03   10.34   clock reconvergence pessimism
                       -0.26   10.08   library setup time
                               10.08   data required time
-----------------------------------------------------------------------------
                               10.08   data required time
                               -7.21   data arrival time
-----------------------------------------------------------------------------
                                2.86   slack (MET)



====================== TT Corner (Typical) ============================

Startpoint: reset (input port clocked by clk)
Endpoint: _412_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical (TT)

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (propagated)
                        2.00    2.00 v input external delay
                0.01    0.00    2.00 v reset (in)
     1    0.00                        reset (net)
                0.01    0.00    2.00 v input2/A (sky130_fd_sc_hd__buf_1)
                0.05    0.10    2.10 v input2/X (sky130_fd_sc_hd__buf_1)
     1    0.01                        net2 (net)
                0.05    0.00    2.10 v _365_/A (sky130_fd_sc_hd__buf_8)
                0.08    0.18    2.28 v _365_/X (sky130_fd_sc_hd__buf_8)
    14    0.10                        _203_ (net)
                0.08    0.01    2.28 v _366_/A (sky130_fd_sc_hd__buf_8)
                0.07    0.19    2.47 v _366_/X (sky130_fd_sc_hd__buf_8)
    16    0.09                        _204_ (net)
                0.08    0.00    2.48 v _367_/A (sky130_fd_sc_hd__inv_2)
                0.03    0.06    2.53 ^ _367_/Y (sky130_fd_sc_hd__inv_2)
     1    0.00                        _005_ (net)
                0.03    0.00    2.53 ^ _412_/SET_B (sky130_fd_sc_hd__dfstp_1)
                        2.53    data arrival time

                       10.00   10.00   clock clk (rise edge)
                        0.00   10.00   clock source latency
                0.07    0.04   10.04 ^ clk (in)
     1    0.01                        clk (net)
                0.07    0.00   10.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.06    0.14   10.19 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                        clknet_0_clk (net)
                0.06    0.00   10.19 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.05    0.13   10.32 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                        clknet_2_3__leaf_clk (net)
                0.05    0.00   10.32 ^ _412_/CLK (sky130_fd_sc_hd__dfstp_1)
                       -0.25   10.07   clock uncertainty
                        0.00   10.07   clock reconvergence pessimism
                        0.14   10.21   library recovery time
                               10.21   data required time
-----------------------------------------------------------------------------
                               10.21   data required time
                               -2.53   data arrival time
-----------------------------------------------------------------------------
                                7.68   slack (MET)


Startpoint: _436_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _446_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical (TT)
```

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.07    0.05    0.05 ^ clk (in)
     1    0.01                           clk (net)
                   0.07    0.00    0.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)

                   0.06    0.16    0.21 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.06    0.00    0.21 ^ clkbuf_2_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.06    0.16    0.36 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
    13    0.04                           clknet_2_0__leaf_clk (net)
                   0.06    0.00    0.36 ^ _436_/CLK (sky130_fd_sc_hd__dfrtp_2)
                   0.07    0.45    0.82 v _436_/Q (sky130_fd_sc_hd__dfrtp_2)
     6    0.02                           RX.count[9] (net)
                   0.07    0.00    0.82 v _223_/A (sky130_fd_sc_hd__or4_1)
                   0.12    0.61    1.42 v _223_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _102_ (net)
                   0.12    0.00    1.42 v _225_/A3 (sky130_fd_sc_hd__o41a_1)
                   0.12    0.48    1.90 v _225_/X (sky130_fd_sc_hd__o41a_1)
     4    0.01                           _104_ (net)
                   0.12    0.00    1.90 v _245_/B (sky130_fd_sc_hd__or4_1)
                   0.12    0.59    2.50 v _245_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _120_ (net)
                   0.12    0.00    2.50 v _252_/A (sky130_fd_sc_hd__or3b_1)
                   0.07    0.42    2.92 v _252_/X (sky130_fd_sc_hd__or3b_1)
     1    0.00                           _125_ (net)
                   0.07    0.00    2.92 v _253_/S (sky130_fd_sc_hd__mux2_1)
                   0.05    0.32    3.24 v _253_/X (sky130_fd_sc_hd__mux2_1)
     1    0.00                           _126_ (net)
                   0.05    0.00    3.24 v _254_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.02    0.09    3.34 v _254_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           _062_ (net)
                   0.02    0.00    3.34 v _446_/D (sky130_fd_sc_hd__dfrtp_1)
                           3.34    3.34   data arrival time

                          10.00   10.00   clock clk (rise edge)
                           0.00   10.00   clock source latency
                   0.07    0.04   10.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.07    0.00   10.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.06    0.14   10.19 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.06    0.00   10.19 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.05    0.13   10.32 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                           clknet_2_3__leaf_clk (net)
                   0.05    0.00   10.32 ^ _446_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          -0.25   10.07   clock uncertainty
                           0.02   10.09   clock reconvergence pessimism
                          -0.10    9.99   library setup time
                                   9.99   data required time
---------------------------------------------------------------------------
                                   9.99   data required time
                                  -3.34   data arrival time
---------------------------------------------------------------------------
                                   6.65   slack (MET)



===========================================================================
report_checks --slack_max -0.01
===========================================================================
====================== FF Corner (Fastest) ==============================

No paths found.

====================== SS Corner (Slowest) ==============================

No paths found.

====================== TT Corner (Typical) ==============================

No paths found.


===========================================================================
 report_check_types -max_slew -max_cap -max_fanout -violators
===========================================================================
====================== FF Corner (Fastest) ==============================


====================== SS Corner (Slowest) ==============================


====================== TT Corner (Typical) ==============================



===========================================================================
report_parasitic_annotation -report_unannotated
```

```
===========================================================================
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

===========================================================================
max slew violation count 0
max fanout violation count 0
max cap violation count 0
===========================================================================

===========================================================================

check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -generated_clocks
===========================================================================
================================================
SOURCE REPORT: designs/uart_bt_car/runs/fix3/reports/signoff/26-mca/rcx_min_sta.power.rpt
================================================

===========================================================================
 report_power
===========================================================================
======================== FF Corner (Fastest) ==============================

Group                  Internal Switching     Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             2.69e-04   8.10e-05   2.13e-07   3.50e-04 44.4%
Combinational          2.59e-04   1.79e-04   2.64e-09   4.38e-04 55.6%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  5.28e-04   2.60e-04   2.15e-07   7.88e-04 100.0%
                          67.0%      33.0%       0.0%

======================== SS Corner (Slowest) ==============================

Group                  Internal Switching     Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             1.80e-04   5.26e-05   8.48e-07   2.34e-04 43.5%
Combinational          1.87e-04   1.15e-04   1.18e-06   3.04e-04 56.5%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  3.67e-04   1.68e-04   2.03e-06   5.37e-04 100.0%
                          68.4%      31.3%       0.4%

======================== TT Corner (Typical) ==============================

Group                  Internal Switching     Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             2.32e-04   6.76e-05   2.12e-07   3.00e-04 44.1%
Combinational          2.30e-04   1.50e-04   1.58e-09   3.80e-04 55.9%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  4.62e-04   2.17e-04   2.14e-07   6.80e-04 100.0%
                          68.0%      31.9%       0.0%

================================================
SOURCE REPORT: designs/uart_bt_car/runs/fix3/reports/signoff/26-mca/rcx_min_sta.summary.rpt
================================================

===========================================================================
report_tns
===========================================================================
tns 0.00

===========================================================================
report_wns
===========================================================================
wns 0.00

===========================================================================
report_worst_slack -max (Setup)
===========================================================================
worst slack 2.86

===========================================================================
report_worst_slack -min (Hold)
===========================================================================
worst slack 0.10
================================================
SOURCE REPORT: designs/uart_bt_car/runs/fix3/reports/signoff/28-mca/rcx_max_sta.checks.rpt
================================================

===========================================================================
report_checks -unconstrained
===========================================================================
======================== FF Corner (Fastest) ==============================

Startpoint: reset (input port clocked by clk)
```

```
Endpoint: _412_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest (FF)

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          2.00    2.00 v input external delay
                   0.01   0.00    2.00 v reset (in)
     1    0.00                           reset (net)
                   0.01   0.00    2.00 v input2/A (sky130_fd_sc_hd__buf_1)
                   0.03   0.06    2.07 v input2/X (sky130_fd_sc_hd__buf_1)

     1    0.01                           net2 (net)
                   0.03   0.00    2.07 v _365_/A (sky130_fd_sc_hd__buf_8)
                   0.06   0.11    2.18 v _365_/X (sky130_fd_sc_hd__buf_8)
    14    0.10                           _203_ (net)
                   0.07   0.02    2.20 v _366_/A (sky130_fd_sc_hd__buf_8)
                   0.06   0.13    2.33 v _366_/X (sky130_fd_sc_hd__buf_8)
    16    0.10                           _204_ (net)
                   0.06   0.01    2.34 v _367_/A (sky130_fd_sc_hd__inv_2)
                   0.03   0.04    2.38 ^ _367_/Y (sky130_fd_sc_hd__inv_2)
     1    0.00                           _005_ (net)
                   0.03   0.00    2.38 ^ _412_/SET_B (sky130_fd_sc_hd__dfstp_1)
                          2.38    2.38   data arrival time

                         10.00   10.00   clock clk (rise edge)
                          0.00   10.00   clock source latency
                   0.06   0.04   10.04 ^ clk (in)
     1    0.02                           clk (net)
                   0.06   0.00   10.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.05   0.10   10.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.05   0.00   10.14 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.09   10.23 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                           clknet_2_3__leaf_clk (net)
                   0.04   0.00   10.23 ^ _412_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25    9.98   clock uncertainty
                          0.00    9.98   clock reconvergence pessimism
                          0.09   10.07   library recovery time
                                 10.07   data required time
-----------------------------------------------------------------------------
                                 10.07   data required time
                                 -2.38   data arrival time
-----------------------------------------------------------------------------
                                  7.68   slack (MET)


Startpoint: _454_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: motor_left_b (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest (FF)

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.06   0.04    0.04 ^ clk (in)
     1    0.02                           clk (net)
                   0.06   0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.05   0.11    0.15 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.05   0.00    0.15 ^ clkbuf_2_2__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.05   0.11    0.26 ^ clkbuf_2_2__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
    13    0.04                           clknet_2_2__leaf_clk (net)
                   0.05   0.00    0.26 ^ _454_/CLK (sky130_fd_sc_hd__dfrtp_1)
                   0.10   0.26    0.53 ^ _454_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.01                           net4 (net)
                   0.10   0.00    0.53 ^ output4/A (sky130_fd_sc_hd__clkbuf_4)
                   0.09   0.15    0.68 ^ output4/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           motor_left_b (net)
                   0.09   0.00    0.68 ^ motor_left_b (out)
                          0.68    0.68   data arrival time

                         10.00   10.00   clock clk (rise edge)
                          0.00   10.00   clock network delay (propagated)
                         -0.25    9.75   clock uncertainty
                          0.00    9.75   clock reconvergence pessimism
                         -2.00    7.75   output external delay
                                 7.75    data required time
-----------------------------------------------------------------------------
                                 7.75    data required time
                                -0.68    data arrival time
-----------------------------------------------------------------------------
                                 7.07    slack (MET)



========================= SS Corner (Slowest) ==============================
```

```
Startpoint: reset (input port clocked by clk)
Endpoint: _412_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest (SS)

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          2.00    2.00 v input external delay
                   0.02   0.01    2.01 v reset (in)
     1    0.00                           reset (net)
                   0.02   0.00    2.01 v input2/A (sky130_fd_sc_hd__buf_1)
                   0.09   0.18    2.19 v input2/X (sky130_fd_sc_hd__buf_1)
     1    0.01                           net2 (net)
                   0.09   0.00    2.19 v _365_/A (sky130_fd_sc_hd__buf_8)

                   0.14   0.32    2.51 v _365_/X (sky130_fd_sc_hd__buf_8)
    14    0.10                           _203_ (net)
                   0.15   0.02    2.53 v _366_/A (sky130_fd_sc_hd__buf_8)
                   0.13   0.35    2.88 v _366_/X (sky130_fd_sc_hd__buf_8)
    16    0.10                           _204_ (net)
                   0.14   0.01    2.89 v _367_/A (sky130_fd_sc_hd__inv_2)
                   0.05   0.10    2.99 ^ _367_/Y (sky130_fd_sc_hd__inv_2)
     1    0.00                           _005_ (net)
                   0.05   0.00    2.99 ^ _412_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  2.99   data arrival time

                         10.00   10.00   clock clk (rise edge)
                          0.00   10.00   clock source latency
                   0.12   0.07   10.07 ^ clk (in)
     1    0.02                           clk (net)
                   0.12   0.00   10.07 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.10   0.26   10.33 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.10   0.00   10.34 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.08   0.24   10.57 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                           clknet_2_3__leaf_clk (net)
                   0.08   0.00   10.58 ^ _412_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   10.33   clock uncertainty
                          0.00   10.33   clock reconvergence pessimism
                          0.27   10.60   library recovery time
                                 10.60   data required time
---------------------------------------------------------------------------
                                 10.60   data required time
                                 -2.99   data arrival time
---------------------------------------------------------------------------
                                  7.61   slack (MET)


Startpoint: _436_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _446_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest (SS)

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.12   0.08    0.08 ^ clk (in)
     1    0.02                           clk (net)
                   0.12   0.00    0.08 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.10   0.29    0.37 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.10   0.00    0.37 ^ clkbuf_2_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.10   0.28    0.65 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
    13    0.04                           clknet_2_0__leaf_clk (net)
                   0.10   0.00    0.65 ^ _436_/CLK (sky130_fd_sc_hd__dfrtp_2)
                   0.15   0.95    1.60 v _436_/Q (sky130_fd_sc_hd__dfrtp_2)
     6    0.02                           RX.count[9] (net)
                   0.15   0.00    1.60 v _223_/A (sky130_fd_sc_hd__or4_1)
                   0.26   1.42    3.03 v _223_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _102_ (net)
                   0.26   0.00    3.03 v _225_/A3 (sky130_fd_sc_hd__o41a_1)
                   0.24   0.97    3.99 v _225_/X (sky130_fd_sc_hd__o41a_1)
     4    0.02                           _104_ (net)
                   0.24   0.00    3.99 v _245_/B (sky130_fd_sc_hd__or4_1)
                   0.24   1.40    5.39 v _245_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _120_ (net)
                   0.24   0.00    5.39 v _252_/A (sky130_fd_sc_hd__or3b_1)
                   0.15   0.98    6.37 v _252_/X (sky130_fd_sc_hd__or3b_1)
     1    0.00                           _125_ (net)
                   0.15   0.00    6.37 v _253_/S (sky130_fd_sc_hd__mux2_1)
                   0.11   0.75    7.12 v _253_/X (sky130_fd_sc_hd__mux2_1)
     1    0.00                           _126_ (net)
                   0.11   0.00    7.12 v _254_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.05   0.18    7.30 v _254_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           _062_ (net)
                   0.05   0.00    7.30 v _446_/D (sky130_fd_sc_hd__dfrtp_1)
                                  7.30   data arrival time
```

```
                        10.00   10.00   clock clk (rise edge)
                         0.00   10.00   clock source latency
                 0.12    0.07   10.07 ^ clk (in)
     1    0.02                         clk (net)
                 0.12    0.00   10.07 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.10    0.26   10.33 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                         clknet_0_clk (net)
                 0.10    0.00   10.34 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.08    0.24   10.57 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                         clknet_2_3__leaf_clk (net)
                 0.08    0.00   10.58 ^ _446_/CLK (sky130_fd_sc_hd__dfrtp_1)
                        -0.25   10.33   clock uncertainty
                         0.04   10.36   clock reconvergence pessimism
                        -0.26   10.10   library setup time
                                10.10   data required time
-----------------------------------------------------------------------------
                                10.10   data required time
                                -7.30   data arrival time
-----------------------------------------------------------------------------
                                 2.80   slack (MET)

====================== TT Corner (Typical) ==============================

Startpoint: reset (input port clocked by clk)
Endpoint: _412_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical (TT)

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock network delay (propagated)
                         2.00    2.00 v input external delay
                 0.01    0.01    2.01 v reset (in)
     1    0.00                         reset (net)
                 0.01    0.00    2.01 v input2/A (sky130_fd_sc_hd__buf_1)
                 0.05    0.10    2.10 v input2/X (sky130_fd_sc_hd__buf_1)
     1    0.01                         net2 (net)
                 0.05    0.00    2.10 v _365_/A (sky130_fd_sc_hd__buf_8)
                 0.08    0.17    2.28 v _365_/X (sky130_fd_sc_hd__buf_8)
    14    0.10                         _203_ (net)
                 0.09    0.02    2.29 v _366_/A (sky130_fd_sc_hd__buf_8)
                 0.08    0.19    2.48 v _366_/X (sky130_fd_sc_hd__buf_8)
    16    0.10                         _204_ (net)
                 0.08    0.01    2.50 v _367_/A (sky130_fd_sc_hd__inv_2)
                 0.03    0.06    2.56 ^ _367_/Y (sky130_fd_sc_hd__inv_2)
     1    0.00                         _005_ (net)
                 0.03    0.00    2.56 ^ _412_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                 2.56   data arrival time

                        10.00   10.00   clock clk (rise edge)
                         0.00   10.00   clock source latency
                 0.07    0.05   10.05 ^ clk (in)
     1    0.02                         clk (net)
                 0.07    0.00   10.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.06    0.15   10.19 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                         clknet_0_clk (net)
                 0.06    0.00   10.20 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.05    0.14   10.33 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                         clknet_2_3__leaf_clk (net)
                 0.05    0.00   10.33 ^ _412_/CLK (sky130_fd_sc_hd__dfstp_1)
                        -0.25   10.08   clock uncertainty
                         0.00   10.08   clock reconvergence pessimism
                         0.14   10.22   library recovery time
                                10.22   data required time
-----------------------------------------------------------------------------
                                10.22   data required time
                                -2.56   data arrival time
-----------------------------------------------------------------------------
                                 7.67   slack (MET)


Startpoint: _436_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _446_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical (TT)

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
                 0.07    0.05    0.05 ^ clk (in)
     1    0.02                         clk (net)
                 0.07    0.00    0.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.06    0.16    0.21 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                         clknet_0_clk (net)
                 0.06    0.00    0.22 ^ clkbuf_2_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.06    0.16    0.38 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
    13    0.05                         clknet_2_0__leaf_clk (net)
                 0.07    0.00    0.38 ^ _436_/CLK (sky130_fd_sc_hd__dfrtp_2)
                 0.07    0.46    0.84 v _436_/Q (sky130_fd_sc_hd__dfrtp_2)
```

```
     6    0.02                           RX.count[9] (net)
                  0.07    0.00    0.84 v _223_/A (sky130_fd_sc_hd__or4_1)
                  0.13    0.61    1.45 v _223_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _102_ (net)
                  0.13    0.00    1.45 v _225_/A3 (sky130_fd_sc_hd__o41a_1)
                  0.13    0.49    1.94 v _225_/X (sky130_fd_sc_hd__o41a_1)
     4    0.02                           _104_ (net)
                  0.13    0.00    1.94 v _245_/B (sky130_fd_sc_hd__or4_1)
                  0.12    0.60    2.54 v _245_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _120_ (net)
                  0.12    0.00    2.54 v _252_/A (sky130_fd_sc_hd__or3b_1)
                  0.07    0.42    2.96 v _252_/X (sky130_fd_sc_hd__or3b_1)
     1    0.00                           _125_ (net)
                  0.07    0.00    2.96 v _253_/S (sky130_fd_sc_hd__mux2_1)
                  0.05    0.33    3.29 v _253_/X (sky130_fd_sc_hd__mux2_1)
     1    0.00                           _126_ (net)
                  0.05    0.00    3.29 v _254_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.02    0.09    3.38 v _254_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           _062_ (net)
                  0.02    0.00    3.38 v _446_/D (sky130_fd_sc_hd__dfrtp_1)

                                  3.38   data arrival time

                         10.00   10.00   clock clk (rise edge)
                          0.00   10.00   clock source latency
                  0.07    0.05   10.05 ^ clk (in)
     1    0.02                           clk (net)
                  0.07    0.00   10.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.06    0.15   10.19 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                  0.06    0.00   10.20 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.14   10.33 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                           clknet_2_3__leaf_clk (net)
                  0.05    0.00   10.33 ^ _446_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   10.08   clock uncertainty
                          0.02   10.10   clock reconvergence pessimism
                         -0.10   10.00   library setup time
                                 10.00   data required time
-----------------------------------------------------------------------------
                                 10.00   data required time
                                 -3.38   data arrival time
-----------------------------------------------------------------------------
                                  6.62   slack (MET)



================================================================================
report_checks --slack_max -0.01
================================================================================
====================== FF Corner (Fastest) ==============================

No paths found.

====================== SS Corner (Slowest) ==============================

No paths found.

====================== TT Corner (Typical) ==============================

No paths found.


================================================================================
 report_check_types -max_slew -max_cap -max_fanout -violators
================================================================================
====================== FF Corner (Fastest) ==============================


====================== SS Corner (Slowest) ==============================


====================== TT Corner (Typical) ==============================



================================================================================
report_parasitic_annotation -report_unannotated
================================================================================
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

================================================================================
max slew violation count 0
max fanout violation count 0
max cap violation count 0
================================================================================

================================================================================
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -generated_clocks
================================================================================
================================================
SOURCE REPORT: designs/uart_bt_car/runs/fix3/reports/signoff/28-mca/rcx_max_sta.power.rpt
```

```
================================================

===========================================================================
 report_power
===========================================================================
====================== FF Corner (Fastest) ============================

Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             2.68e-04   8.62e-05   2.13e-07   3.54e-04 44.1%
Combinational          2.60e-04   1.89e-04   2.64e-09   4.49e-04 55.9%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  5.28e-04   2.75e-04   2.15e-07   8.03e-04 100.0%
                          65.7%      34.3%       0.0%

====================== SS Corner (Slowest) ============================

Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------

Sequential             1.80e-04   5.61e-05   8.48e-07   2.36e-04 43.2%
Combinational          1.87e-04   1.22e-04   1.18e-06   3.11e-04 56.8%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  3.67e-04   1.78e-04   2.03e-06   5.47e-04 100.0%
                          67.1%      32.6%       0.4%

====================== TT Corner (Typical) ============================

Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             2.31e-04   7.20e-05   2.12e-07   3.04e-04 43.9%
Combinational          2.30e-04   1.58e-04   1.58e-09   3.88e-04 56.1%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  4.62e-04   2.30e-04   2.14e-07   6.92e-04 100.0%
                          66.7%      33.2%       0.0%

================================================
SOURCE REPORT: designs/uart_bt_car/runs/fix3/reports/signoff/28-mca/rcx_max_sta.summary.rpt
================================================

===========================================================================
report_tns
===========================================================================
tns 0.00

===========================================================================
report_wns
===========================================================================
wns 0.00

===========================================================================
report_worst_slack -max (Setup)
===========================================================================
worst slack 2.80

===========================================================================
report_worst_slack -min (Hold)
===========================================================================
worst slack 0.11
================================================
SOURCE REPORT: designs/uart_bt_car/runs/fix3/reports/signoff/30-mca/rcx_nom_sta.checks.rpt
================================================

===========================================================================
report_checks -unconstrained
===========================================================================
====================== FF Corner (Fastest) ============================

Startpoint: reset (input port clocked by clk)
Endpoint: _412_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest (FF)

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          2.00    2.00 v input external delay
                   0.01   0.00    2.00 v reset (in)
     1    0.00                           reset (net)
                   0.01   0.00    2.00 v input2/A (sky130_fd_sc_hd__buf_1)
                   0.03   0.06    2.07 v input2/X (sky130_fd_sc_hd__buf_1)
     1    0.01                           net2 (net)
```

```
                0.03    0.00    2.07 v _365_/A (sky130_fd_sc_hd__buf_8)
                0.06    0.12    2.18 v _365_/X (sky130_fd_sc_hd__buf_8)
   14    0.10                           _203_ (net)
                0.06    0.01    2.19 v _366_/A (sky130_fd_sc_hd__buf_8)
                0.06    0.13    2.32 v _366_/X (sky130_fd_sc_hd__buf_8)
   16    0.10                           _204_ (net)
                0.06    0.01    2.33 v _367_/A (sky130_fd_sc_hd__inv_2)
                0.03    0.04    2.37 ^ _367_/Y (sky130_fd_sc_hd__inv_2)
    1    0.00                           _005_ (net)
                0.03    0.00    2.37 ^ _412_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                2.37   data arrival time

                       10.00   10.00   clock clk (rise edge)
                        0.00   10.00   clock source latency
                0.05    0.03   10.03 ^ clk (in)
    1    0.01                           clk (net)
                0.05    0.00   10.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.05    0.10   10.13 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
    4    0.04                           clknet_0_clk (net)
                0.05    0.00   10.13 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.04    0.09   10.23 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
    9    0.03                           clknet_2_3__leaf_clk (net)
                0.04    0.00   10.23 ^ _412_/CLK (sky130_fd_sc_hd__dfstp_1)
                       -0.25    9.98   clock uncertainty
                        0.00    9.98   clock reconvergence pessimism
                        0.09   10.06   library recovery time

                               10.06   data required time
-----------------------------------------------------------------------------
                               10.06   data required time
                               -2.37   data arrival time
-----------------------------------------------------------------------------
                                7.69   slack (MET)


Startpoint: _454_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: motor_left_b (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest (FF)

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock source latency
                0.05    0.04    0.04 ^ clk (in)
    1    0.01                           clk (net)
                0.05    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.05    0.11    0.15 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
    4    0.04                           clknet_0_clk (net)
                0.05    0.00    0.15 ^ clkbuf_2_2__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.05    0.11    0.26 ^ clkbuf_2_2__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
   13    0.04                           clknet_2_2__leaf_clk (net)
                0.05    0.00    0.26 ^ _454_/CLK (sky130_fd_sc_hd__dfrtp_1)
                0.06    0.27    0.53 v _454_/Q (sky130_fd_sc_hd__dfrtp_1)
    2    0.01                           net4 (net)
                0.06    0.00    0.53 v output4/A (sky130_fd_sc_hd__clkbuf_4)
                0.06    0.14    0.67 v output4/X (sky130_fd_sc_hd__clkbuf_4)
    1    0.03                           motor_left_b (net)
                0.06    0.00    0.67 v motor_left_b (out)
                                0.67   data arrival time

                       10.00   10.00   clock clk (rise edge)
                        0.00   10.00   clock network delay (propagated)
                       -0.25    9.75   clock uncertainty
                        0.00    9.75   clock reconvergence pessimism
                       -2.00    7.75   output external delay
                                7.75   data required time
-----------------------------------------------------------------------------
                                7.75   data required time
                               -0.67   data arrival time
-----------------------------------------------------------------------------
                                7.08   slack (MET)



======================== SS Corner (Slowest) ==============================

Startpoint: reset (input port clocked by clk)
Endpoint: _412_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest (SS)

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (propagated)
                        2.00    2.00 v input external delay
                0.02    0.01    2.01 v reset (in)
    1    0.00                           reset (net)
                0.02    0.00    2.01 v input2/A (sky130_fd_sc_hd__buf_1)
                0.09    0.18    2.19 v input2/X (sky130_fd_sc_hd__buf_1)
```

```
     1    0.01                           net2 (net)
                  0.09    0.00    2.19 v _365_/A (sky130_fd_sc_hd__buf_8)
                  0.14    0.32    2.51 v _365_/X (sky130_fd_sc_hd__buf_8)
    14    0.10                           _203_ (net)
                  0.14    0.01    2.52 v _366_/A (sky130_fd_sc_hd__buf_8)
                  0.13    0.34    2.87 v _366_/X (sky130_fd_sc_hd__buf_8)
    16    0.10                           _204_ (net)
                  0.13    0.01    2.87 v _367_/A (sky130_fd_sc_hd__inv_2)
                  0.05    0.10    2.97 ^ _367_/Y (sky130_fd_sc_hd__inv_2)
     1    0.00                           _005_ (net)
                  0.05    0.00    2.97 ^ _412_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  2.97   data arrival time

                         10.00   10.00   clock clk (rise edge)
                          0.00   10.00   clock source latency
                  0.11    0.07   10.07 ^ clk (in)
     1    0.01                           clk (net)
                  0.11    0.00   10.07 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.10    0.26   10.33 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                  0.10    0.00   10.33 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.08    0.24   10.56 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                           clknet_2_3__leaf_clk (net)
                  0.08    0.00   10.56 ^ _412_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   10.31   clock uncertainty
                          0.00   10.31   clock reconvergence pessimism
                          0.27   10.59   library recovery time
                                 10.59   data required time
-----------------------------------------------------------------------------

                                 10.59   data required time
                                 -2.97   data arrival time
-----------------------------------------------------------------------------
                                  7.62   slack (MET)


Startpoint: _436_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _446_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest (SS)

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.11    0.08    0.08 ^ clk (in)
     1    0.01                           clk (net)
                  0.11    0.00    0.08 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.10    0.28    0.36 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                  0.10    0.00    0.36 ^ clkbuf_2_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.10    0.28    0.64 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
    13    0.04                           clknet_2_0__leaf_clk (net)
                  0.10    0.00    0.64 ^ _436_/CLK (sky130_fd_sc_hd__dfrtp_2)
                  0.14    0.94    1.58 v _436_/Q (sky130_fd_sc_hd__dfrtp_2)
     6    0.02                           RX.count[9] (net)
                  0.14    0.00    1.59 v _223_/A (sky130_fd_sc_hd__or4_1)
                  0.25    1.42    3.00 v _223_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _102_ (net)
                  0.25    0.00    3.00 v _225_/A3 (sky130_fd_sc_hd__o41a_1)
                  0.23    0.96    3.96 v _225_/X (sky130_fd_sc_hd__o41a_1)
     4    0.02                           _104_ (net)
                  0.23    0.00    3.96 v _245_/B (sky130_fd_sc_hd__or4_1)
                  0.24    1.39    5.35 v _245_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _120_ (net)
                  0.24    0.00    5.35 v _252_/A (sky130_fd_sc_hd__or3b_1)
                  0.15    0.98    6.33 v _252_/X (sky130_fd_sc_hd__or3b_1)
     1    0.00                           _125_ (net)
                  0.15    0.00    6.33 v _253_/S (sky130_fd_sc_hd__mux2_1)
                  0.11    0.75    7.07 v _253_/X (sky130_fd_sc_hd__mux2_1)
     1    0.00                           _126_ (net)
                  0.11    0.00    7.07 v _254_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.18    7.25 v _254_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           _062_ (net)
                  0.05    0.00    7.25 v _446_/D (sky130_fd_sc_hd__dfrtp_1)
                                  7.25   data arrival time

                         10.00   10.00   clock clk (rise edge)
                          0.00   10.00   clock source latency
                  0.11    0.07   10.07 ^ clk (in)
     1    0.01                           clk (net)
                  0.11    0.00   10.07 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.10    0.26   10.33 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                  0.10    0.00   10.33 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.08    0.24   10.56 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                           clknet_2_3__leaf_clk (net)
                  0.08    0.00   10.57 ^ _446_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   10.32   clock uncertainty
                          0.03   10.35   clock reconvergence pessimism
                         -0.26   10.09   library setup time
                                 10.09   data required time
```

```
                                10.09   data required time
                                -7.25   data arrival time
-------------------------------------------------------------------------------
                                 2.83   slack (MET)


====================== TT Corner (Typical) ==============================

Startpoint: reset (input port clocked by clk)
Endpoint: _412_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical (TT)

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          2.00    2.00 v input external delay
                   0.01   0.01    2.01 v reset (in)
     1    0.00                           reset (net)
                   0.01   0.00    2.01 v input2/A (sky130_fd_sc_hd__buf_1)
                   0.05   0.10    2.10 v input2/X (sky130_fd_sc_hd__buf_1)
     1    0.01                           net2 (net)
                   0.05   0.00    2.10 v _365_/A (sky130_fd_sc_hd__buf_8)
                   0.08   0.18    2.28 v _365_/X (sky130_fd_sc_hd__buf_8)
    14    0.10                           _203_ (net)
                   0.08   0.01    2.29 v _366_/A (sky130_fd_sc_hd__buf_8)
                   0.08   0.19    2.48 v _366_/X (sky130_fd_sc_hd__buf_8)
    16    0.10                           _204_ (net)

                   0.08   0.01    2.48 v _367_/A (sky130_fd_sc_hd__inv_2)
                   0.03   0.06    2.54 ^ _367_/Y (sky130_fd_sc_hd__inv_2)
     1    0.00                           _005_ (net)
                   0.03   0.00    2.54 ^ _412_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  2.54   data arrival time

                         10.00   10.00   clock clk (rise edge)
                          0.00   10.00   clock source latency
                   0.07   0.05   10.05 ^ clk (in)
     1    0.01                           clk (net)
                   0.07   0.00   10.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.06   0.14   10.19 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.06   0.00   10.19 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.05   0.13   10.33 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                           clknet_2_3__leaf_clk (net)
                   0.05   0.00   10.33 ^ _412_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   10.08   clock uncertainty
                          0.00   10.08   clock reconvergence pessimism
                          0.14   10.22   library recovery time
                                 10.22   data required time
-------------------------------------------------------------------------------
                                 10.22   data required time
                                 -2.54   data arrival time
-------------------------------------------------------------------------------
                                  7.67   slack (MET)


Startpoint: _436_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _446_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical (TT)

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.07   0.05    0.05 ^ clk (in)
     1    0.01                           clk (net)
                   0.07   0.00    0.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.06   0.16    0.21 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                   0.06   0.00    0.21 ^ clkbuf_2_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.06   0.16    0.37 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
    13    0.04                           clknet_2_0__leaf_clk (net)
                   0.06   0.00    0.37 ^ _436_/CLK (sky130_fd_sc_hd__dfrtp_2)
                   0.07   0.46    0.83 v _436_/Q (sky130_fd_sc_hd__dfrtp_2)
     6    0.02                           RX.count[9] (net)
                   0.07   0.00    0.83 v _223_/A (sky130_fd_sc_hd__or4_1)
                   0.13   0.61    1.44 v _223_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _102_ (net)
                   0.13   0.00    1.44 v _225_/A3 (sky130_fd_sc_hd__o41a_1)
                   0.12   0.49    1.92 v _225_/X (sky130_fd_sc_hd__o41a_1)
     4    0.02                           _104_ (net)
                   0.12   0.00    1.92 v _245_/B (sky130_fd_sc_hd__or4_1)
                   0.12   0.60    2.52 v _245_/X (sky130_fd_sc_hd__or4_1)
     3    0.01                           _120_ (net)
                   0.12   0.00    2.52 v _252_/A (sky130_fd_sc_hd__or3b_1)
                   0.07   0.42    2.94 v _252_/X (sky130_fd_sc_hd__or3b_1)
```

```
     1    0.00                           _125_ (net)
                  0.07    0.00    2.94 v _253_/S (sky130_fd_sc_hd__mux2_1)
                  0.05    0.33    3.27 v _253_/X (sky130_fd_sc_hd__mux2_1)
     1    0.00                           _126_ (net)
                  0.05    0.00    3.27 v _254_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.02    0.09    3.36 v _254_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           _062_ (net)
                  0.02    0.00    3.36 v _446_/D (sky130_fd_sc_hd__dfrtp_1)
                                  3.36   data arrival time

                         10.00   10.00   clock clk (rise edge)
                          0.00   10.00   clock source latency
                  0.07    0.05   10.05 ^ clk (in)
     1    0.01                           clk (net)
                  0.07    0.00   10.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.06    0.14   10.19 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     4    0.04                           clknet_0_clk (net)
                  0.06    0.00   10.19 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.13   10.33 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
     9    0.03                           clknet_2_3__leaf_clk (net)
                  0.05    0.00   10.33 ^ _446_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   10.08   clock uncertainty
                          0.02   10.10   clock reconvergence pessimism
                         -0.10    9.99   library setup time
                                  9.99   data required time
-----------------------------------------------------------------------------
                                  9.99   data required time
                                 -3.36   data arrival time
-----------------------------------------------------------------------------
                                  6.63   slack (MET)



================================================================================

report_checks --slack_max -0.01
================================================================================
====================== FF Corner (Fastest) ================================

No paths found.

====================== SS Corner (Slowest) ================================

No paths found.

====================== TT Corner (Typical) ================================

No paths found.


================================================================================
 report_check_types -max_slew -max_cap -max_fanout -violators
================================================================================
====================== FF Corner (Fastest) ================================


====================== SS Corner (Slowest) ================================


====================== TT Corner (Typical) ================================



================================================================================
report_parasitic_annotation -report_unannotated
================================================================================
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

================================================================================
max slew violation count 0
max fanout violation count 0
max cap violation count 0
================================================================================

================================================================================
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -generated_clocks
================================================================================
================================================
SOURCE REPORT: designs/uart_bt_car/runs/fix3/reports/signoff/30-mca/rcx_nom_sta.power.rpt
================================================

================================================================================
 report_power
================================================================================
====================== FF Corner (Fastest) ================================

Group                  Internal Switching     Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             2.68e-04   8.37e-05   2.13e-07   3.52e-04 44.3%
Combinational          2.60e-04   1.84e-04   2.64e-09   4.43e-04 55.7%
```

```
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
------------------------------------------------------------------
Total                  5.28e-04   2.68e-04   2.15e-07   7.96e-04 100.0%
                          66.3%      33.6%       0.0%

======================= SS Corner (Slowest) ==============================

Group                  Internal Switching     Leakage      Total
                          Power      Power      Power      Power (Watts)
------------------------------------------------------------------
Sequential             1.80e-04   5.44e-05   8.48e-07   2.35e-04 43.3%
Combinational          1.87e-04   1.19e-04   1.18e-06   3.07e-04 56.7%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
------------------------------------------------------------------
Total                  3.67e-04   1.73e-04   2.03e-06   5.43e-04 100.0%
                          67.7%      32.0%       0.4%

======================= TT Corner (Typical) ==============================

Group                  Internal Switching     Leakage      Total
                          Power      Power      Power      Power (Watts)
------------------------------------------------------------------
Sequential             2.32e-04   7.00e-05   2.12e-07   3.02e-04 44.0%
Combinational          2.30e-04   1.54e-04   1.58e-09   3.84e-04 56.0%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
------------------------------------------------------------------
Total                  4.62e-04   2.24e-04   2.14e-07   6.86e-04 100.0%
                          67.3%      32.6%       0.0%

================================================
SOURCE REPORT: designs/uart_bt_car/runs/fix3/reports/signoff/30-mca/rcx_nom_sta.summary.rpt
================================================

===========================================================================
report_tns

===========================================================================
tns 0.00

===========================================================================
report_wns
===========================================================================
wns 0.00

===========================================================================
report_worst_slack -max (Setup)
===========================================================================
worst slack 2.83

===========================================================================
report_worst_slack -min (Hold)
===========================================================================
worst slack 0.11
```